

University of Botswana
Department of Computer Science
Tutorial 4 - Mathematical Induction

Question 1

Prove by mathematical induction that

$$2^n > n^2 \quad \forall n \geq 5$$

Question 2

Prove by mathematical induction that

$$n^3 > 2n + 1 \quad \forall n \geq 2$$

Question 3

Prove by mathematical induction that $n^3 - 7n + 3$ is divisible by 3 $\forall n \geq 1$. in class.

Question 4

Prove by mathematical induction that

$$1 + 2 + 4 + 8 + \dots + 2^{n-1} = 2^n - 1 \quad \forall n \geq 1.$$

Question 5

Prove by mathematical induction that $23^n - 1$ is divisible by 11, $\forall n \geq 0$.

Question 6

Prove by mathematical induction that

$$1 + 3 + 5 + \dots + (2n - 1) = n^2, \quad \forall n \geq 1$$

Question 7

Prove by mathematical induction that, for all $n \geq 1$.

$$1^3 + 2^3 + 3^3 + 4^3 + \dots + n^3 = \frac{n^2(n+1)^2}{4}$$

Question 8

Prove by mathematical induction that, for all $n \geq 1$.

$$1^2 - 2^2 + 3^2 - \dots + (-1)^{n-1} n^2 = \frac{(-1)^{n-1} n(n+1)}{2}$$